BACHELOR'S THESIS

Modelling Wellbeing: Comparing Statistical, Machine Learning, Hybrid and Bayesian Network Models on Psychological Data

Table of Contents

Chapter 1 - Introduction

1.1. Motivation

Mental health and psychological wellbeing are among the most complex and consequential outcomes in modern society. Conditions such as depression, chronic stress, and low life satisfaction affect hundreds of millions of people worldwide, with significant consequences for individuals, organisations, and public health systems. Despite a broad scientific consensus on the factors that contribute to wellbeing - including social support, economic security, physical health, and psychological resilience - the question of how accurately and interpretably these factors can be modelled using computational methods remains open.

Two parallel challenges motivate this thesis. The first is the distinction between individual-level and societal-level data. Survey responses from employees or students capture rich personal experiences but are characterised by high noise, subjective bias, and complex interactions among predictors. Aggregate indicators such as GDP per capita, life expectancy, and institutional trust capture structural conditions at the country level but abstract away individual variation. Whether predictive models perform differently across these two levels of analysis - and whether the same modeling frameworks can be applied consistently to both - is a question with both methodological and practical implications.

The second challenge concerns the tension between prediction and explanation. Machine learning models have demonstrated strong predictive performance across a wide range of psychological and health outcomes, but their internal logic is often opaque. Statistical models offer interpretability but may miss nonlinear relationships. Bayesian Networks occupy a distinctive position: they are generative probabilistic models that can encode causal structure explicitly, enabling reasoning under uncertainty in a way that neither regression nor ensemble methods can match. Whether Bayesian Networks can recover plausible causal structures from observational wellbeing data - and whether those structures add value beyond what machine learning feature importance already reveals - is the central methodological question of this thesis [Pearl2009, Scutari2014].

These challenges are not purely academic. Policymakers designing mental health interventions, employers evaluating workplace support programs, and researchers studying the determinants of happiness all need models that are not only accurate but interpretable and causally grounded. A model that predicts depression with high accuracy but cannot explain which factors drive the prediction, or in which direction, is of limited practical value for intervention design [Shmueli2010, Breiman2001].

1.2. Scope

This thesis presents a systematic comparative study of statistical, machine learning, hybrid, and Bayesian Network models for predicting and explaining psychological wellbeing under uncertainty, applied consistently across both individual-level and societal-level data.

The study is grounded in four datasets spanning two levels of analysis. At the individual level, three datasets are examined: a survey of mental health conditions in the technology industry, a survey of depression among students, and a Canadian population health survey measuring life satisfaction. At the societal level, the World Happiness Report Gallup Poll dataset - covering 149 countries across 2005–2020 - provides a macro-level comparison. An additional synthetic Kaggle dataset nominally derived from the World Happiness Report is included as a data quality case study, demonstrating the consequences of absent signal for all modelling approaches.

The scope is deliberately comparative. The same modeling frameworks are applied consistently across all datasets, enabling direct cross-dataset and cross-model evaluation. The study does not aim to perform clinical diagnosis or psychological assessment. Its focus is on modelling self-reported mental health and wellbeing outcomes using different predictive approaches, with particular emphasis on the causal structure of the data and the interpretability of model outputs.

Wellbeing is operationalised across multiple constructs depending on the dataset: work interference and treatment-seeking behaviour in the workplace context [Diener1984], depression as a binary clinical indicator in the student context, life satisfaction as a continuous self-report measure in the Canadian survey [Pavot1993], and the Cantril Life Ladder score as a societal happiness measure in the Gallup dataset [Helliwell2023, OECD2020].

1.3. Research Questions

The study is organised around five research questions that together address the comparative performance, interpretability, causal validity, and robustness of the modelling frameworks examined:

RQ1. How do different modeling approaches perform when predicting wellbeing at individual versus societal levels? This question examines whether predictive performance - measured by classification metrics at the individual level and regression metrics at the societal level - differs systematically between data levels, and whether that difference varies across model families.

RQ2. How does interpretability differ between regression, machine learning, and Bayesian Networks? This question examines the qualitative and quantitative differences in how each model family explains its predictions, using coefficient analysis, SHAP values, feature importance rankings, and Bayesian Network conditional probability tables as complementary interpretability tools [Lundberg2017, Molnar2019, Goldstein2015].

RQ3. Are Bayesian Networks better at capturing plausible causal structures in wellbeing data? This question evaluates whether Bayesian Network structure learning, with and without expert constraints, recovers directed acyclic graphs whose edge directions are consistent with established domain knowledge about the causal determinants of mental health and wellbeing [Pearl2009, Morgan2015].

RQ4. Can Bayesian Networks reveal causal structures that ML models cannot? This question examines whether the explicit directional edges learned by Bayesian Networks provide causal information beyond what machine learning feature importance conveys, and what the conditions are under which data-driven structure learning succeeds or fails [Yarkoni2017, Scutari2014].

RQ5. How robust are these models to missing data, noise, and correlated predictors? This question evaluates model stability across cross-validation folds, sensitivity to the removal of dominant predictors, and performance under conditions of high missingness and correlated feature sets [Hastie2009, Kuhn2013].

1.4. Overview of Approach

The methodology follows a consistent pipeline applied to all datasets. Each dataset is preprocessed in two versions: a machine learning version with imputation, one-hot encoding, and missingness indicators, and a Bayesian Network version preserving missing values as discrete states for structure learning. Baseline machine learning models - logistic or linear regression, decision tree, random forest, and XGBoost - are trained and evaluated using stratified cross-validation with multiple metrics. Hybrid models combining nonlinear feature selection with linear estimators, and stacking ensembles, are evaluated against the baselines. Bayesian Networks are learned using Hill-Climb Search with BIC scoring, both unconstrained and with expert-defined forbidden edges encoding domain knowledge about causal ordering.

Interpretability is analysed at two levels. At the global level, SHAP TreeExplainer values provide model-agnostic feature importance and directional effect estimates for the best-performing model on each dataset. Sensitivity analyses remove dominant proxy predictors to assess robustness. At the structural level, Bayesian Network graphs are inspected for causally implausible edges, and BIC scores are compared between unconstrained and constrained versions to quantify the statistical cost of expert constraints.

Cross-dataset comparison is performed through unified performance tables, visualisations of hybrid model gain over baselines, and a Bayesian Network consolidation summarising structure quality, inference results, and causal pathway robustness across all datasets. The thesis concludes with a discussion connecting empirical findings to the five research questions and to the broader prediction-explanation debate in psychological science [Breiman2001, Shmueli2010].

Chapter 2 – Literature Review

2.1. Overview

This chapter situates the thesis within three intersecting bodies of literature: the debate between predictive and explanatory modelling in statistical and psychological science, the theory and practice of Bayesian Network causal discovery, and the measurement of subjective wellbeing at individual and societal levels. Together these three strands provide the conceptual foundation for the comparative modelling framework developed in this thesis and motivate the five research questions introduced in Chapter 1.

2.2. Prediction vs Explanation Debate

The relationship between statistical explanation and predictive accuracy has been a central tension in quantitative science for decades. The distinction was formalised most sharply by Breiman [Breiman2001], who identified two cultures in statistical modelling: the data modelling culture, which assumes a stochastic data-generating process and prioritises interpretable parametric models, and the algorithmic modelling culture, which treats the process as unknown and prioritises predictive accuracy. Breiman argued that psychology and the social sciences had become overly committed to the data modelling culture at the expense of predictive validity, and that algorithmic methods such as ensemble trees and neural networks offered substantial gains in out-of-sample performance.

Shmueli [Shmueli2010] developed this distinction more formally by introducing a framework for distinguishing explanatory modelling - aimed at testing causal hypotheses about the data-generating process - from predictive modelling - aimed at producing accurate forecasts of new observations. She argued that the two goals require different model selection criteria, different evaluation metrics, and different interpretive standards, and that conflating them leads to both poor predictions and poor causal inference. A model that fits the training data well and has theoretically grounded coefficients may nevertheless perform poorly on held-out data, while a model with strong out-of-sample performance may be causally uninterpretable. This framework directly motivates the comparative design of this thesis, which evaluates both predictive performance and causal interpretability across model families.

Yarkoni and Westfall [Yarkoni2017] extended this argument specifically to psychological science, demonstrating that psychology's near-total focus on explaining behaviour through controlled experiments had produced a literature of intricate causal theories with little predictive validity. They proposed that adopting machine learning principles - in particular regularisation, cross-validation, and out-of-sample evaluation - would produce more accurate and generalisable psychological models. Their argument is directly relevant to the wellbeing prediction context: the variables that best explain depression or life satisfaction in a theoretical framework may not be the same as those that best predict it in a held-out sample.

The practical implication of this debate for the present thesis is that no single modelling approach is optimal across both goals simultaneously. Logistic and linear regression models offer coefficient-based explanations grounded in theory but may miss nonlinear relationships. Ensemble methods such as Random Forest and XGBoost offer strong predictive performance but require post-hoc interpretability tools. Bayesian Networks offer explicit causal structure but may sacrifice predictive accuracy in exchange for interpretability. Understanding this trade-off empirically - across multiple datasets and outcome types - is the central contribution of the present work.

The growing adoption of complex machine learning models in applied settings has generated a parallel literature on post-hoc interpretability methods. Hastie, Tibshirani, and Friedman [Hastie2009] provide the statistical foundation for tree-based and ensemble methods, including the conditions under which feature importance measures are reliable. Friedman [Friedman2001] introduced gradient boosting machines and partial dependence plots as tools for understanding the marginal effect of predictors in nonlinear models.

Goldstein and colleagues [Goldstein2015] introduced individual conditional expectation plots as a refinement of partial dependence plots, enabling visualisation of heterogeneous predictor effects across individuals rather than only average marginal effects. Lundberg and Lee [Lundberg2017] unified feature importance methods under the SHAP framework, providing a theoretically grounded decomposition of model predictions into additive feature contributions based on Shapley values from cooperative game theory. SHAP values satisfy desirable axiomatic properties - local accuracy, consistency, and missingness - that earlier importance measures do not, making them the standard tool for explaining individual predictions from tree-based models. Molnar [Molnar2019] provides a comprehensive overview of the interpretable machine learning landscape, situating

SHAP alongside partial dependence plots, permutation importance, and local surrogate models within a unified taxonomy of explanation methods.

In this thesis, SHAP is applied as the primary interpretability tool for the best-performing ML model on each dataset, complementing the structural interpretability provided by Bayesian Network graphs and coefficient tables from linear models.

2.3. Bayesian Networks Causal Discovery

Bayesian Networks are directed acyclic graphical models that represent a joint probability distribution over a set of variables through a product of conditional distributions [Scutari2014]. Each node represents a variable and each directed edge represents a conditional dependency, encoding the assertion that a child variable is directly influenced by its parent variables given the remaining network structure. The absence of an edge between two nodes encodes conditional independence, making Bayesian Networks a natural language for expressing causal hypotheses about data-generating processes.

Pearl [Pearl2009] established the theoretical foundations for causal inference in graphical models, introducing the do-calculus as a formal framework for reasoning about interventions - the effect of setting a variable to a fixed value - as distinct from observations. His framework distinguishes between associational quantities, which can be estimated from observational data, and causal quantities, which require either experimental manipulation or structural assumptions encoded in a directed acyclic graph. The d-separation criterion provides a graphical test for conditional independence, enabling the derivation of testable implications from a causal model without intervention data.

Morgan and Winship [Morgan2015] situate Bayesian Networks and graphical causal models within the broader counterfactual tradition of causal inference, connecting Pearl's do-calculus to the potential outcomes framework of Rubin and Neyman. They discuss the conditions under which observational data can support causal claims, emphasising the role of structural assumptions - encoded as forbidden or required edges in a Bayesian Network - in making identification possible.

Structure learning algorithms attempt to recover the graph structure of a Bayesian Network from observational data. Score-based methods such as Hill-Climb Search evaluate candidate graphs using a scoring function - typically the Bayesian Information Criterion - and search for the graph that maximises the score [Scutari2014]. The BIC penalises model complexity relative to sample size, favouring sparser graphs when data is limited. Constraint-based methods such as the PC algorithm use conditional independence tests to identify edges, while hybrid methods combine both approaches.

A fundamental limitation of data-driven structure learning is that it cannot distinguish between Markov equivalent graphs - directed acyclic graphs that encode the same conditional independence relationships but differ in edge direction. Without additional constraints derived from domain knowledge - such as the temporal ordering of variables or theoretical assumptions about causal direction - structure learning algorithms may recover statistically equivalent but causally implausible graphs. This limitation is directly relevant to the wellbeing context, where survey variables often measure co-occurring constructs whose causal ordering is theoretically motivated but not identifiable from observational data alone.

In this thesis, expert knowledge is incorporated into the structure learning process by specifying forbidden edges encoding known causal ordering constraints - for example, that demographic variables such as age and gender cannot be caused by psychological outcome variables, and that treatment-seeking behaviour cannot cause a family history of mental illness. The impact of these constraints on both statistical fit and causal plausibility is evaluated systematically across all datasets.

2.4. Wellbeing Measurement

Subjective wellbeing is a multidimensional construct encompassing cognitive evaluations of life quality, the presence of positive affect, and the absence of negative affect [Diener1984]. Diener's foundational review established subjective wellbeing as a legitimate scientific construct and identified its three primary components: life satisfaction as a cognitive judgement, positive affect as the frequency of pleasant emotional states, and negative affect as the frequency of unpleasant emotional states. This tripartite model has been the dominant framework in wellbeing research for four decades and provides the theoretical grounding for the outcome variables used in this thesis.

At the individual level, life satisfaction is most commonly measured using the Satisfaction With Life Scale [Pavot1993], a five-item instrument with established psychometric properties including high internal consistency, temporal stability, and convergent validity with related constructs. The review by Pavot and Diener confirms the scale's validity across diverse populations and its sensitivity to changes in life circumstances, making it appropriate for modelling as a continuous outcome variable. In the Canadian Survey dataset used in this thesis, life satisfaction is measured as a single-item numeric rating on a 0–10 scale, which is consistent with established single-item life satisfaction measures used in large-scale population surveys.

At the societal level, the World Happiness Report provides the most comprehensive annual assessment of national happiness across countries [Helliwell2023]. The primary measure is the Cantril Self-Anchoring Striving Scale - referred to as the Life Ladder - in which respondents rate their current life on a scale from 0 (worst possible life) to 10 (best possible life). The WHR framework identifies six explanatory factors that account for variation in national happiness scores: GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption. These factors are precisely the predictors available in the Gallup dataset analysed in this thesis, enabling a direct empirical test of whether their importance rankings recovered by machine learning and Bayesian Network models are consistent with the established WHR framework.

The OECD [OECD2020] provides a complementary framework for measuring societal wellbeing that extends beyond happiness to include material conditions, quality of life, sustainability, and inequality. The OECD guidelines distinguish between objective indicators - income, employment, housing - and subjective indicators - life satisfaction, affect balance, meaning - and emphasise the importance of measuring both dimensions for a complete picture of population wellbeing. This distinction motivates the inclusion of both objective socioeconomic variables (income, food security, employment) and subjective psychological states (mental health, stress, sense of belonging) as predictors in the Canadian Survey models.

Mental health as a specific dimension of individual wellbeing has received increasing attention as both a public health priority and a modelling target. Depression is among the most prevalent mental health conditions globally, affecting an estimated 280 million people according to the World Health Organisation, and is associated with substantial reductions in quality of life, work productivity, and social functioning. The student depression and mental health in tech datasets used in this thesis capture two distinct at-risk populations - students facing academic and financial pressure, and technology workers facing workplace-specific mental health stigma and support access barriers - providing complementary individual-level perspectives on mental health outcomes.

Chapter 3 – Methodology

3.1. Overview

This chapter describes the methodological framework used to address the five research questions set out in Chapter 1. The study adopts a comparative design in which four families of models - statistical regression, machine learning, hybrid, and Bayesian Networks - are applied to five datasets varying in scale, measurement level, and task type. Two datasets involve classification of psychological outcomes at the individual level; two involve regression of life satisfaction at the individual level; and one involves regression of subjective wellbeing at the societal (macro) level. A fifth dataset, synthetically generated, is retained as a data quality case study rather than a substantive modelling exercise. The consistent application of the same model families across all datasets enables direct comparison of predictive performance, interpretability, and structural validity.

Section 3.2 describes each dataset and the preprocessing decisions applied to it. Section 3.3 introduces the four model families and their configurations. Section 3.4 covers the evaluation framework, including metrics, cross-validation strategy, and the handling of the evaluation asymmetry between ML and BN pipelines. Section 3.5 gives a full account of the Bayesian Network structure learning approach, covering the algorithm, scoring function, expert knowledge constraints, parameter estimation, and inference procedure.

3.2. Dataset descriptions and preprocessing decisions

Five datasets were selected to cover a range of wellbeing constructs, measurement instruments, and analytical challenges. The distinction between micro-level (individual survey data) and macro-level (country-aggregated data) is treated as a primary axis of comparison throughout the thesis.

3.2.1 Mental Health in Tech Survey

The OSMI Mental Health in Tech Survey is a publicly available dataset collected via self-report from technology-sector workers. After removing observations with missing target values, the working sample comprised 995 respondents and 43 features. The prediction target was a three-class variable, *combined_target*, constructed by crossing the work interference item (*work_interfere*) with whether the respondent had sought treatment: Class 0 indicates no interference, Class 1 indicates interference without treatment, and Class 2 indicates interference with treatment. The class distribution was approximately 60.6%, 21.4%, and 18.0% respectively.

Preprocessing for the machine learning pipeline included the addition of binary missingness indicators for features with non-trivial missing rates, followed by median imputation for continuous variables and mode imputation for categoricals. Nominal features were one-hot encoded with reference categories dropped to prevent multicollinearity. For the Bayesian Network pipeline, missing values were preserved as a distinct state rather than imputed, ordinal encodings were retained, and high-cardinality geographic features (*Country* and *state*) were excluded, as their large number of discrete states would render conditional probability tables sparse and unstable.

3.2.2 Student Depression Dataset

This dataset contains survey responses from students on academic, social, and psychological variables. After filtering to confirmed student respondents - removing 0.1% of rows identified as non-students - the sample comprised 27,870 observations. The binary target, *Depression_yes*, indicated presence or absence of depression. Four features were dropped prior to modelling: *City*, which contained a mix of city names and degree abbreviations indicating data entry inconsistency; *Profession*, redundant after student filtering; and *Work Pressure* and *Job Satisfaction*, which are not applicable to a student population. The final feature set comprised 23 variables.

A notable characteristic of this dataset is the dominance of the suicidal ideation feature, which accounts for approximately 52–56% of intrinsic variable importance in tree-based models. Because suicidal ideation is a co-symptom of depression rather than an antecedent cause, its concentration of predictive signal raises questions about whether the models are learning genuine risk-factor structure or simply detecting a correlated symptom. This is examined in sensitivity analysis. For the Bayesian Network, age and CGPA were discretised into bins to satisfy the discrete variable requirement; continuous features in the ML pipeline were retained in their original form.

3.2.3 Canadian Community Health Survey

This large-scale national health survey comprises approximately 104,000 respondents. The continuous target, *Life_satisfaction*, is measured on a 0–10 scale. Columns with more than 60% missing values were dropped prior to modelling, removing features such as BMI for the 12–17 age group, fruit and vegetable consumption, and smoking status. The target missingness indicator was excluded from the feature set to prevent leakage. After preprocessing, approximately 58 features were retained.

Two variables - *Mental_health_state* and *Gen_health_state* - are self-rated health variables that function as proxies for life satisfaction: they correlate strongly with the target but arguably reflect a similar evaluative process rather than being independent causal inputs. Their contribution is examined in a dedicated sensitivity analysis. *Food_security* is coded ordinally from 0 (fully food-secure) to 3 (severely insecure). *BMI_18_above* is coded as 1 (underweight or normal) and 2 (overweight or obese).

3.2.4 WHR Synthetic Dataset

This dataset, presented on Kaggle as a World Happiness Report synthetic dataset, contains 4,000 observations across 10 countries and 20 years. The *Happiness_Score* column was dropped prior to modelling due to direct target leakage. Initial modelling revealed that all model families returned R^2 values near zero, and the maximum absolute correlation between any feature and the target was 0.049. The dataset contains no genuine predictive signal and is therefore not used as a substantive modelling exercise. It is retained as a case study in data quality detection, illustrating how a consistent modelling pipeline can identify a null-signal dataset before interpretability or causal analysis is attempted. The methodological implications of this finding are discussed in Section 4.4.1.

3.2.5 WHR Gallup Dataset

The Gallup World Poll data underlying the World Happiness Report was obtained via Kaggle and represents the most credible macro-level dataset in this study. The working sample comprised 1,949 observations across 149 countries and the period 2005–2020. The target variable, *Life_Ladder*, is the Cantril self-anchoring scale, which asks respondents to place their current life on a ladder from 0 (worst possible life) to 10 (best possible life). Seventeen features were retained after preprocessing, including log GDP per capita, social support, healthy life expectancy at birth, freedom to make life choices, generosity, perceptions of corruption, positive affect, negative affect, year, and binary missingness indicators for all columns with missing values. The maximum absolute feature-target correlation was 0.783, confirming genuine predictive signal. For the Bayesian Network, all continuous variables were quantile-discretised into five equal-frequency bins; year was binned into five-year periods to reduce the number of discrete states while preserving temporal ordering.

3.3. Model families and configurations

Four model families are applied consistently across all datasets. Each family represents a distinct position on the trade-off between predictive performance and interpretability, and their comparison is central to Research Questions 1, 2, and 3.

3.3.1 Statistical Baseline Models

The statistical baseline models serve as the interpretability anchor for the study. For classification tasks, logistic regression is used; for regression tasks, ordinary least squares linear regression is used. These models are selected not because they are expected to achieve the highest predictive accuracy, but because they are maximally transparent: every predictor's effect is directly readable from its estimated coefficient, including direction and magnitude. This transparency makes them the natural reference point for the interpretability comparisons developed in later chapters.

A single decision tree (classifier or regressor depending on task) is also included in the baseline set. Decision trees occupy an intermediate position: they are non-parametric and capable of capturing non-linear relationships and interactions, yet their structure can be visualised and traversed directly. Maximum depth is constrained via cross-validation to limit overfitting.

3.3.2 Machine Learning Models

Two ensemble methods form the primary machine learning models. Random Forest constructs a large ensemble of decision trees, each trained on a bootstrap sample of the training data with a random subset of features considered at each split. Predictions are aggregated by majority vote (classification) or averaging (regression). The randomisation introduced at both the sample and feature level reduces variance relative to any individual tree, and the ensemble's feature importance scores - based on the aggregate reduction in node impurity across all trees - provide a global measure of variable relevance.

XGBoost implements gradient boosting, constructing trees sequentially such that each new tree is fitted to the pseudo-residuals of the current ensemble under a differentiable loss function.

Regularisation is applied through a learning rate (shrinkage), subsampling of rows and columns, and L1/L2 penalties on leaf weights. XGBoost achieves the highest or near-highest predictive performance across all datasets in this study. Its predictions are used as the basis for the SHAP interpretability analysis described in Section 3.4.

Both models are evaluated using five-fold cross-validation with stratification on the target variable. Hyperparameters are held at well-established defaults throughout; the primary goal is comparison across model families rather than maximum optimisation of any individual model.

3.3.3 Hybrid Models

Hybrid models combine the feature selection capacity of ensemble methods with the interpretability of linear models. Two selection-then-estimation pipelines are evaluated: one using Random Forest importance scores to select features, followed by logistic or linear regression on the reduced feature set; and one using XGBoost importance scores in the same role. A third hybrid uses stacking, training a logistic or linear regression meta-learner on the out-of-fold predictions generated by Random Forest and XGBoost as base learners.

The inclusion of hybrid models reflects a practically motivated question: whether the interpretability of a linear model can be recovered without substantially sacrificing the predictive power of ensemble methods. The cross-dataset finding that hybrid models yield negligible gains over the best ensemble baseline is itself a substantive result, discussed in Chapter 5.

3.4. Evaluation framework

Evaluation metrics are chosen to match the task type and to reflect the priorities established in the research questions.

For classification tasks - MH Tech and Student Depression - the primary metric is macro-averaged F1 score, which computes precision and recall separately for each class before averaging, weighting all classes equally regardless of their frequency in the data. This is appropriate given the class imbalance present in both datasets, where a majority-class classifier would score misleadingly well on accuracy. Area under the ROC curve (AUC) is reported as a threshold-independent complement; for the three-class MH Tech target, AUC is computed using a one-vs-rest strategy.

For regression tasks - Canadian Survey, WHR Gallup, and WHR Synthetic - the primary metric is R^2, the proportion of variance in the target explained by the model. Root mean squared error (RMSE) is reported alongside R^2 to provide a metric expressed in the original scale of the target variable, which aids interpretability of effect magnitudes.

All machine learning models are evaluated using stratified five-fold cross-validation, and the reported metrics are the mean values across the five held-out folds. Cross-validated metrics are the primary basis for performance comparisons, as they provide an estimate of out-of-sample generalisation rather than in-sample fit. Bayesian Network predictions are computed on the full dataset because the pgmpy inference pipeline does not readily support fold-level re-estimation of structure and parameters. This evaluation asymmetry - CV metrics for ML models versus full-data metrics for BNs - is acknowledged throughout the results and discussion chapters. Where direct numerical comparisons are made between ML and BN performance, this asymmetry is explicitly noted, as BN full-data metrics are expected to be modestly optimistic relative to cross-validated equivalents.

3.5. BN structure learning approach

Bayesian Networks are probabilistic graphical models that represent the joint distribution of a set of random variables as a directed acyclic graph (DAG). Each node in the graph represents a variable; each directed edge from node A to node B encodes a direct conditional dependence of B on A, with B's distribution specified as a conditional probability table (CPT) over the states of its parents. The full joint distribution factorises as the product of these local conditional distributions, which makes inference computationally tractable and the learned structure directly interpretable as a set of conditional independence claims.

All Bayesian Networks in this study are implemented using pgmpy version 1.1.0.

3.5.1 Structure Learning Algorithm

Structure learning - the task of identifying the DAG from data - is performed using the Hill Climb Search (HCS) algorithm. HCS begins with an empty graph and iteratively proposes single-edge operations: addition, removal, or reversal of a directed edge. At each step, the operation yielding the greatest improvement in a scoring function is accepted; the algorithm terminates when no single-edge operation improves the score. HCS is a score-based greedy search and does not guarantee a globally optimal solution, but it is computationally practical for the variable counts encountered in this study and has been shown to recover plausible structures in comparable applied settings (Scutari and Denis, 2014).

To prevent overfitting through excessive parent sets, the maximum in-degree - the number of parents permitted for any single node - is constrained to three. This limit reduces the combinatorial search space, prevents the learning of highly specific conditional distributions that would not generalise, and produces structures that are more readily interpretable.

3.5.2 Scoring Function

The scoring function used throughout is BIC-d, the Bayesian Information Criterion adapted for discrete data. For a DAG G and dataset D of n observations, the BIC score is:

$$\mathrm{BIC}(G, D) = \log P(D \mid G, \hat{\theta}) - (k / 2) \cdot \log n$$

where $\log P(D \mid G, \hat{\theta})$ is the log-likelihood of the data under the maximum likelihood parameter estimates $\hat{\theta}$, and k is the total number of free parameters across all CPTs in the graph. The penalty term $(k / 2) \cdot \log n$ discourages the addition of edges whose likelihood gain does not outweigh the cost of the additional parameters they introduce. HCS maximises this score, meaning structures with more edges are only preferred when they provide a sufficiently better fit to justify their complexity.

The pgmpy 1.1.0 built-in BIC scorer is incompatible with the BDeu parameter estimates used in this pipeline due to internal inconsistencies in how the library computes sufficient statistics for mixed prior and ML estimation. BIC scores are therefore computed using a custom implementation that directly evaluates the log-likelihood of the data under the learned CPTs and subtracts the standard BIC penalty. This implementation was validated against manually computed scores on small example graphs prior to use.

3.5.3 Expert Knowledge Constraints

An unconstrained Hill Climb Search will sometimes recover edges that are statistically supported by the data but domain-implausible - for example, an outcome variable appearing as a parent of a demographic variable, or a time-invariant characteristic being caused by a time-varying one. To quantify this problem, an unconstrained BN is estimated for each dataset and all recovered edges are evaluated against domain knowledge. Implausible edges are catalogued and the implausibility rate (number of implausible edges as a proportion of total edges) is reported.

A restricted BN is then estimated by encoding domain constraints as a forbidden edges list, passed to the HCS algorithm via pgmpy's ExpertKnowledge interface. Forbidden edges prevent the algorithm from ever placing a directed edge in a specified direction between a specified pair of nodes during the search. The categories of forbidden edges applied in this study include: demographic or background variables as targets of outcome or symptom variables; time-invariant variables as targets of time-varying ones; and any edge whose direction would imply a causal pathway that is biologically, temporally, or theoretically impossible given the survey design.

For the Canadian Survey, where the proxy status of *Mental_health_state* and *Gen_health_state* is a concern, an additional strongly restricted BN is estimated with those variables removed entirely from the structure search, producing a three-way comparison: unconstrained, restricted, and strongly restricted.

The BIC cost of applying constraints is defined as the percentage change in BIC score between the unconstrained and restricted models:

BIC cost (%) = ((BIC_restricted − BIC_unconstrained) / |BIC_unconstrained|) × 100

A positive value indicates the restricted model achieves a better (higher) BIC score than the unconstrained model - a result that occurs when the unconstrained search has overfit to noise via implausible edges. A negative value indicates a fitness cost: the constraints force the algorithm away from statistically preferred structures. Reporting this cost across datasets provides a quantitative characterisation of how much the unconstrained algorithm's output conflicts with domain knowledge on each dataset.

3.5.4 Parameter Estimation

Conditional probability tables are estimated using the BayesianEstimator with a BDeu (Bayesian Dirichlet equivalent uniform) prior. BDeu assigns a uniform prior over the parameters of each CPT, distributed equally across all parent configurations, with the total prior weight controlled by the equivalent sample size hyperparameter. An equivalent sample size of 10 is used throughout, which provides mild smoothing of sparse cells without substantially distorting CPTs that are well-supported by data. This is particularly important for the MH Tech and WHR Gallup datasets, where some combinations of parent states have few observations in the sample.

3.5.5 Inference and Prediction

Probabilistic inference is performed using Variable Elimination (VE). VE is an exact inference algorithm that computes the posterior distribution of a query variable given observed evidence by systematically eliminating non-query, non-evidence variables through marginalisation. For a query variable with evidence e, VE returns the full conditional distribution P(target | e), from which the most probable state (for classification) or the expected value computed from bin midpoints (for regression) is used as the point prediction.

For regression targets that have been quantile-discretised into bins, the expected value prediction is:

$$\hat{E}[\text{target} \mid e] = \Sigma\_i \text{ midpoint\_i} \cdot P(\text{target} = \text{bin\_i} \mid e)$$

where the sum is taken over all bins and midpoint_i is the midpoint of the original continuous range corresponding to that bin. This converts the BN's categorical posterior into a scalar prediction that can be evaluated using standard regression metrics.

Chapter 4 - Results

4.1. Mental Health in Tech

4.1.1 Baseline ML Results

The four baseline models were evaluated on the three-class *combined_target* using five-fold stratified cross-validation. Results are reported in Table 4.1.

Table 4.1. MH Tech - Baseline ML cross-validated performance.

TODO table

Model	CV Macro F1	CV AUC
Logistic Regression	0.457	0.685
Decision Tree	0.408	0.640
Random Forest	0.470	0.696
XGBoost	0.441	0.691

Random Forest achieved the highest macro F1 (0.470) and AUC (0.696) among the baseline models, with logistic regression and XGBoost performing comparably. The decision tree was the weakest baseline on both metrics, consistent with its tendency to overfit on noisy, moderate-sized samples without ensemble averaging.

The overall performance level across all baseline models is modest. Macro F1 values in the range 0.41–0.47 reflect the difficulty of the three-class prediction task, the class imbalance between the majority no-interference class (60.6%) and the two interference classes, and the inherent noise in self-report survey data. The relatively small gap between logistic regression and Random Forest - 0.013 F1 points - suggests that the predictive structure in this dataset is largely captured by linear relationships, with limited additional signal recoverable through non-linear modelling.

4.1.2 Hybrid ML Results

Three hybrid configurations were evaluated: Random Forest feature selection followed by logistic regression (RF -> LogReg), XGBoost feature selection followed by logistic regression (XGB -> LogReg), and stacking with a logistic regression meta-learner. Results are reported in Table 4.2.

Table 4.2. MH Tech - Hybrid ML cross-validated performance.

TODO table

Model CV Macro F1 CV AUC

RF Selection -> LogReg 0.465 0.701

XGB Selection -> LogReg 0.488 0.701

Stacking (LogReg meta) 0.460 0.700

The XGB -> LogReg pipeline achieved the best hybrid F1 (0.488), marginally exceeding the best baseline (Random Forest, 0.470) by 0.018 F1 points. This is the largest hybrid improvement observed across any dataset in this study, yet it remains small in absolute terms. The AUC improvement is negligible across all three hybrid configurations (0.700–0.701 versus 0.696 for the best baseline).

The stacking meta-learner underperformed relative to the selection-then-regression pipelines despite having access to both base learners' predictions, suggesting that the linear meta-learner was unable to extract meaningful additional signal from the combination of Random Forest and XGBoost outputs on this dataset.

4.1.3 BN Results

Two Bayesian Network versions were estimated: an unconstrained BN and a restricted BN incorporating domain-knowledge forbidden edges.

The unconstrained BN learned a graph containing 41 edges. Manual evaluation of these edges against domain knowledge identified 10 as implausible, yielding an implausibility rate of 24%. Examples of implausible edges include demographic outcomes being placed as parents of time-invariant background variables, and treatment-seeking behaviour appearing as a cause of symptom severity in a direction inconsistent with the survey's temporal logic.

The restricted BN was estimated after encoding a forbidden edges list to prevent the algorithm from recovering these implausible structures. The BIC cost of applying these constraints was −3.98%, indicating a modest fitness cost: the restricted model trades a small amount of statistical fit for a structurally coherent graph.

The restricted BN produced the following direct parents for the target nodes:

- Parents of *work_interfere*: Gender, family_history
- Parents of *treatment*: Gender, work_interfere

This structure encodes a two-stage pathway in which gender and family history of mental illness jointly influence the degree to which mental health affects work, and in which both gender and the experience of work interference influence whether an individual seeks treatment. The identification of *family_history* as a direct parent of *work_interfere* is consistent with the well-established heritability of mood disorders and anxiety, and with findings from the SHAP analysis described in Section 4.1.4.

Predictive performance of the restricted BN, evaluated on the full dataset, yielded a macro F1 of 0.471 and an AUC of 0.771. The F1 is comparable to the best baseline ML model (0.470), though the evaluation asymmetry - full-data metrics for the BN versus cross-validated metrics for ML - means this comparison should be interpreted cautiously. The AUC of 0.771 exceeds the ML baselines (maximum 0.701), but this advantage is likely partially attributable to the same evaluation asymmetry rather than to genuine structural superiority.

4.1.4 SHAP Analysis

SHAP values were computed for the Random Forest model, which achieved the highest baseline F1. Because the target is three-class, SHAP values are computed separately for each class using a one-vs-rest decomposition, allowing the directional contribution of each feature to be examined per outcome.

The dominant predictor across all three classes is *family_history*. High values of *family_history* produce a strong positive push toward Class 2 (interference with treatment), and a corresponding negative push away from Class 0 (no interference). This pattern is consistent with the BN finding that *family_history* is a direct parent of *work_interfere* and suggests that familial risk is the single strongest marker of severe, treatment-seeking outcomes in this dataset.

care_options_Yes and *benefits_Yes* - variables indicating whether an employer offers mental health care options and benefits - are consistently among the top predictors across classes. Their directional effects indicate that awareness of available care options is positively associated with treatment-seeking (Class 2), which may reflect either genuine access effects or a selection effect in which more health-aware individuals are more likely both to seek treatment and to be aware of employer provisions.

Age is a secondary predictor with a modest positive association with Class 0, suggesting that older respondents are somewhat more likely to report no work interference. Self-employment status and company size contribute lower but non-negligible SHAP values, consistent with the literature on occupational and organisational factors in help-seeking behaviour.

The overall SHAP profile for MH Tech is characterised by the dominance of a single background variable (*family_history*) over occupational and employer-level features, which has implications for the interpretation of workplace mental health interventions discussed in Chapter 6.

4.2. Student Depression

4.2.1 Baseline ML Results

The four baseline models were evaluated on the binary *Depression_yes* target using five-fold stratified cross-validation. Results are reported in Table 4.3.

Table 4.3. Student Depression - Baseline ML cross-validated performance.

TODO table

Model	CV Macro F1	CV AUC
Logistic Regression	0.841	0.921
Decision Tree	0.822	0.901
Random Forest	0.837	0.917
XGBoost	0.841	0.921

Performance across all baseline models is substantially higher than for MH Tech, with macro F1 values ranging from 0.822 to 0.841. Logistic regression and XGBoost are tied for the highest F1 (0.841) and AUC (0.921). Random Forest performs only marginally below these, while the decision tree again underperforms relative to the ensemble methods.

The strong performance of logistic regression, matching the best ensemble model, indicates that the predictive structure in this dataset is well captured by linear decision boundaries. This is consistent with the concentration of signal in a small number of dominant features - in particular, suicidal ideation - whose effect on the target is large and approximately monotone.

4.2.2 Hybrid ML Results

Hybrid results are reported in Table 4.4.

Table 4.4. Student Depression - Hybrid ML cross-validated performance.

TODO table

Model CV Macro F1 CV AUC

RF Selection -> LogReg 0.839 0.921

XGB Selection -> LogReg 0.840 0.921

Stacking (LogReg meta) 0.840 0.921

No hybrid configuration improves on the best baseline. The maximum hybrid F1 is 0.840, one thousandth below the XGBoost baseline of 0.841, and AUC is identical across all hybrid and baseline configurations at 0.921. This near-complete flatness of performance across model families is unusual and reflects the signal structure of the dataset: when a small number of features dominate prediction to the degree observed here, feature selection and meta-learning introduce negligible additional information.

4.2.3 BN Results

The unconstrained BN learned a graph containing 14 edges, of which 11 were evaluated as implausible against domain knowledge - an implausibility rate of 79%, the highest observed across all five datasets. The concentration of implausible edges in a dataset with a dominant co-symptom predictor is interpretable: the structure learning algorithm detects strong statistical associations between suicidal ideation, depression, and other variables, but encodes them in directions that reverse or misattribute causal priority. For example, depression appears as a parent of academic pressure in the unconstrained graph, a direction that is temporally and theoretically implausible given that academic pressure is an ongoing environmental stressor rather than a consequence of clinical diagnosis.

The restricted BN, estimated after applying the forbidden edges list, yielded a BIC cost of −0.29%, indicating that the domain constraints impose a negligible fitness cost while substantially improving the plausibility of the recovered structure.

The restricted BN identified the following direct parents of *Depression_yes*:

- Parents of *Depression_yes*: Academic Pressure, Financial Stress

This is a parsimonious and theoretically coherent structure. Academic pressure and financial stress are both well-established antecedent risk factors for depression in student populations (Helliwell et al., 2023; Diener, 1984), and their identification as direct parents - rather than suicidal ideation, which is a co-symptom - reflects the value of domain constraints in recovering substantively meaningful structure from data dominated by a proxy variable.

Predictive performance of the restricted BN on the full dataset was macro F1 = 0.826, AUC = 0.906. The gap between the BN and the best ML model (0.841 F1) is 0.015 F1 points, the smallest such gap across all datasets in this study. This near-parity suggests that the parsimonious BN structure - two direct parents - captures the majority of the predictive signal available in the dataset.

4.2.4 SHAP Analysis

SHAP values were computed for XGBoost, which matched logistic regression for the highest baseline F1 while providing richer non-linear structure for SHAP decomposition.

The most important feature by a substantial margin is suicidal ideation, with a mean absolute SHAP value of approximately 1.22. This dwarfs the second and third most important features: Academic Pressure (mean |SHAP| ≈ 0.95) and Financial Stress (mean |SHAP| ≈ 0.65). All directional effects are consistent with theory: high suicidal ideation, high academic pressure, and high financial stress all push predictions toward depression (positive SHAP contributions), while low values of each push predictions away from depression.

The dominance of suicidal ideation in the SHAP profile is important to interpret carefully. Suicidal ideation is not a cause of depression but a correlated symptom, and a model that relies primarily on this feature to achieve high F1 is not learning the kind of risk-factor structure that would be useful for early intervention or preventive design. The SHAP profile therefore reinforces the case for the sensitivity analysis in Section 4.2.5.

Beyond the top three features, CGPA (inversely associated with depression), dietary habits, and sleep duration contribute modest but consistent SHAP values in theoretically expected directions.

4.2.5 Sensitivity Analysis

To assess whether meaningful predictive structure remains once the dominant co-symptom proxy is removed, the full ML pipeline was re-run with suicidal ideation excluded from the feature set.

Table 4.5. Student Depression - Sensitivity analysis (without suicidal ideation).

TODO table

Model	CV Macro F1 (full)	CV Macro F1 (no suicidal ideation)	Δ F1
Logistic Regression	0.841	0.791	−0.050
XGBoost	0.841	0.791	−0.050
CV AUC (XGBoost)	0.921	0.873	−0.048

Removing suicidal ideation reduces macro F1 by 0.050 and AUC by 0.048 across all models. While this represents a meaningful reduction, the residual performance - F1 = 0.791, AUC = 0.873 - remains substantially above chance, confirming that genuine antecedent signal is present in the remaining features. After exclusion, the SHAP importance ranking shifts: Academic Pressure becomes the dominant predictor, followed by Financial Stress and CGPA, an ordering closely aligned with the BN's learned parent structure. This convergence between the BN structural result and the post-exclusion SHAP ranking provides mutually reinforcing evidence that these two stressors are the most substantively important modifiable risk factors in this dataset.

4.1. Canadian Survey

4.3.1 Baseline ML Results

The four baseline models were evaluated on the continuous *Life_satisfaction* target using five-fold cross-validation. Results are reported in Table 4.6.

Table 4.6. Canadian Survey - Baseline ML cross-validated performance.

TODO table

Model	CV R^2	CV RMSE
Linear Regression	0.406	1.300
Decision Tree	0.389	1.319
Random Forest	0.407	1.29
XGBoost	0.422	1.282

XGBoost achieved the best baseline performance (CV R^2 = 0.422, RMSE = 1.282), marginally outperforming Random Forest (R^2 = 0.407). Linear regression performed comparably to Random Forest, again indicating that a substantial proportion of the predictive structure is linear. The decision tree underperformed all other baselines.

An R^2 of 0.422 represents moderate predictive accuracy for a life satisfaction outcome measured at the individual level. Individual-level wellbeing is inherently subject to transient psychological states, measurement error, and idiosyncratic life circumstances that aggregate-level models do not face, and a ceiling on R^2 is therefore expected. The role of the self-rated health proxy variables in inflating this ceiling is examined in Section 4.3.5.

4.3.2 Hybrid ML Results

Results for the three hybrid configurations are reported in Table 4.7.

Table 4.7. Canadian Survey - Hybrid ML cross-validated performance.

TODO table

Model	CV R^2	CV RMSE
RF Selection -> LinReg	0.404	1.303
XGB Selection -> LinReg	0.406	1.301
Stacking (LinReg meta)	0.423	1.281

The stacking configuration marginally outperforms the best baseline (R^2 = 0.423 vs 0.422), a difference of 0.001 R^2 points. The selection-then-regression pipelines produce no improvement, with both falling slightly below the linear regression baseline. This pattern is consistent with the other datasets: feature selection followed by a linear model recovers most but not all of the variance captured by the linear baseline, and stacking provides at most a trivial incremental gain over the best single model.

4.3.3 BN Results

The unconstrained BN learned 85 edges, of which 28 were evaluated as implausible - an implausibility rate of 33%. This is the largest raw count of implausible edges across all datasets, reflecting the large number of variables (approximately 58) and the complex correlation structure of a broad national health survey.

Two restricted BN versions were estimated. The restricted BN applied the standard forbidden edges list encoding temporal, demographic, and causal implausibility constraints. The strongly restricted BN additionally excluded *Mental_health_state* and *Gen_health_state* from the structure search entirely, to assess whether their presence distorts the graph learned for the remaining variables.

The BIC cost of the restricted BN was +0.06% relative to the unconstrained model, indicating a marginal improvement in BIC score - meaning that the constraints removed statistically noisy edges as well as domain-implausible ones. The strongly restricted BN incurred a BIC cost of −4.43%, reflecting the meaningful fitness cost of excluding the two proxy variables that carry substantial shared variance with the target.

Both the restricted and strongly restricted BNs identified the same direct parents for *Life_satisfaction*:

- Parents of *Life_satisfaction*: Musculoskeletal_con, Food_security, Income_source

This structure is notable for several reasons. First, it is stable across both constrained versions, indicating robustness to the inclusion or exclusion of the proxy variables. Second, all three identified parents are modifiable social and economic determinants of health, consistent with the OECD framework for wellbeing measurement (OECD, 2020). Food security and income source represent material conditions of daily life; musculoskeletal conditions represent chronic physical health burden, which has a well-documented negative relationship with subjective wellbeing.

Predictive performance of the restricted BN on the full dataset was $R^2 = 0.076$, RMSE = 1.622; the strongly restricted BN yielded $R^2 = 0.078$, RMSE = 1.620. These values are substantially below the ML baselines (best CV $R^2 = 0.422$), representing the largest predictive gap between BN and ML observed in this study. This gap reflects the cost of constraining the BN to a parsimonious, causally plausible structure: a three-parent model cannot compete with an ensemble of hundreds of trees incorporating 58 features. The interpretive trade-off this represents - structural coherence at the cost of predictive coverage - is a central theme of the cross-dataset analysis in Chapter 5.

4.3.4 SHAP Analysis

SHAP values were computed for XGBoost, the best-performing baseline model.

The two dominant predictors are *Mental_health_state* (mean $|\text{SHAP}| \approx 0.40$) and *Gen_health_state* (mean $|\text{SHAP}| \approx 0.31$). Their combined SHAP contribution accounts for the majority of the model's predictive variance. The directional effects are asymmetric: poor mental and general health depress life satisfaction far more than good health elevates it, a pattern consistent with loss aversion effects documented in subjective wellbeing research (Diener, 1984). This asymmetry is visible in the beeswarm plots, where low health state values produce large negative SHAP contributions while high values cluster near zero.

Beyond the proxy variables, *Stress_level* is the third most important predictor, with higher stress consistently pushing life satisfaction downward. *Sense_belonging* appears as the fourth predictor with a positive directional effect - respondents reporting a stronger sense of community belonging report higher life satisfaction, consistent with social connection theories of wellbeing (Helliwell et al., 2023).

Food security and income source, identified by the BN as direct parents of the target, appear in the SHAP ranking but at lower importance than the proxy variables. This ordering reflects the statistical dominance of the proxies rather than their causal priority, a distinction the BN analysis is designed to address.

4.3.5 Sensitivity Analysis

To assess the contribution of the proxy variables and identify the predictors that remain important in their absence, the full ML pipeline was re-run after removing *Mental_health_state* and *Gen_health_state*.

Table 4.8. Canadian Survey - Sensitivity analysis (without proxy variables).

TODO table

Model	CV R^2 (full)	CV R^2 (no proxies)	ΔR^2
Linear Regression	0.406	0.298	−0.108
XGBoost	0.422	0.314	−0.108

Removing the proxy variables reduces CV R^2 by 0.108 across both models - the largest sensitivity-induced drop observed in this study. This confirms that the two self-rated health variables account for approximately a quarter of the total explained variance. The residual R^2 of 0.314 is non-trivial and indicates that genuine predictive structure remains in the remaining features.

After exclusion, *Stress_level* becomes the dominant predictor in the SHAP ranking, followed by *Sense_belonging* and *Food_security*. The emergence of food security as a top-three predictor in the proxy-excluded model is particularly notable: it converges with the BN's identification of food security as a direct parent of life satisfaction, providing cross-method evidence that material food access is a genuine upstream determinant rather than a correlate of the proxy variables.

4.1. World Happiness Report

4.4.1 Synthetic Dataset - Null Result and Methodological Discussion

The WHR synthetic dataset presented an anomalous result at the first stage of modelling: all model families returned R^2 values near zero, and no hybrid or BN configuration recovered any meaningful predictive structure. The maximum absolute correlation between any feature and the target *Life_Satisfaction* was 0.049, consistent with a dataset in which the target variable is effectively independent of all available predictors.

The Bayesian Network analysis confirmed this diagnosis: both the unconstrained and restricted BN structures learned zero edges, indicating that the scoring function found no feature-target association strong enough to justify even a single directed connection.

This outcome demonstrates a valuable property of the modelling pipeline adopted in this study: when applied to a null-signal dataset, it produces a coherent null result rather than an artefactually inflated one. A pipeline susceptible to overfitting would produce positive R^2 values on training data even in the absence of genuine signal; the cross-validated evaluation framework correctly returns near-zero performance estimates, and the BN's failure to learn any structure provides a complementary structural confirmation.

The methodological implication is that the WHR synthetic dataset cannot serve as a basis for claims about societal wellbeing predictors, and that its Kaggle presentation as a real-world dataset constitutes a data quality failure. The detection of this failure before any interpretability or causal analysis was attempted illustrates the importance of beginning comparative modelling studies with baseline signal verification. No further analysis of this dataset is reported.

4.4.2 Gallup Dataset - Results

4.4.2.1 Baseline ML Results

The four baseline models were evaluated on the continuous *Life_Ladder* target using five-fold cross-validation. Results are reported in Table 4.9.

Table 4.9. WHR Gallup - Baseline ML cross-validated performance.

TODO table

Model	CV R^2	CV RMSE
Linear Regression	0.763	0.543
Decision Tree	0.785	0.516
Random Forest	0.843	0.442
XGBoost	0.850	0.432

The WHR Gallup dataset yields the strongest predictive performance of any dataset in this study, with XGBoost achieving CV $R^2 = 0.850$ and RMSE = 0.432. Random Forest is close behind ($R^2 = 0.843$). The decision tree outperforms linear regression on this dataset - a reversal of the pattern seen in the micro-level datasets - suggesting that the relationship between societal indicators and life satisfaction contains meaningful non-linearities and interactions that a single tree is better positioned to capture than a global linear model.

The high R^2 values relative to the Canadian Survey (best baseline 0.422) reflect the fundamental difference between macro- and micro-level prediction. Country-level aggregates of GDP, social support, and institutional quality are structurally cleaner predictors of average national life satisfaction than individual-level survey responses are of any given person's satisfaction, as idiosyncratic noise is averaged away in the aggregation process.

4.4.2.2 Hybrid ML Results

Results for the hybrid configurations are reported in Table 4.10.

Table 4.10. WHR Gallup - Hybrid ML cross-validated performance.

TODO table

Model	CV R^2	CV RMSE
RF Selection -> LinReg	0.759	0.547
XGB Selection -> LinReg	0.755	0.551
Stacking (LinReg meta)	0.840	0.446

The hybrid models underperform the best baseline across all configurations. The selection-then-regression pipelines drop to $R^2 \approx 0.757$, a decline of approximately 0.09 R^2 points relative to the XGBoost baseline, indicating that forcing the non-linear ensemble's selected features into a linear model loses substantial predictive information on this dataset. Stacking recovers most of the gap ($R^2 = 0.840$) but still falls short of the XGBoost baseline (0.850). The WHR Gallup dataset is the only case in this study where hybrid models produce a performance decline rather than a neutral result, underscoring that the non-linear structure of macro-level societal data is not well served by linear meta-learners.

4.4.2.3 BN Results

The unconstrained BN learned 9 edges, of which 1 was evaluated as implausible - an implausibility rate of 11%, the lowest across all datasets. This result is notable: on the dataset with the highest data quality and strongest genuine signal, the unsupervised structure learning algorithm recovers a largely plausible graph without requiring domain correction. This suggests that the BN's tendency to learn implausible edges is amplified by noisy, high-dimensional survey data and attenuated when the underlying relationships are strong and consistent.

The restricted BN, estimated after applying the forbidden edges list to correct the single implausible edge, achieved a BIC cost of +8.77% - the best BIC improvement observed across all datasets. The positive sign indicates that the constraints actually improved the BIC score, meaning the single implausible edge in the unconstrained graph was statistically harmful as well as theoretically implausible, and its removal improved fit.

The restricted BN learned the following structure of direct relevance to the target:

- Parent of *Life_Ladder*: Log_GDP_per_capita (sole direct parent)
- Broader structure: Generosity -> Health -> GDP -> Life_Ladder

This is a clean, interpretable causal chain. GDP per capita is the sole direct predictor of national life satisfaction in the BN's learned structure, with generosity and healthy life expectancy identified as upstream contributors to GDP. The structure is consistent with the economic and sociological literature on national wellbeing (Helliwell et al., 2023) and with the Easterlin tradition linking aggregate income to life satisfaction at the cross-national level.

Predictive performance of the restricted BN on the full dataset was $R^2 = 0.613$, RMSE = 0.880. The gap between the BN (0.613) and the best ML model (CV $R^2 = 0.850$) is 0.237 R^2 points -

substantial, but smaller than the gap observed for the Canadian Survey in proportional terms. The BN achieves this performance using a single direct parent for the target, whereas XGBoost uses all 17 features with full non-linear interactions. The structural parsimony of the BN comes at a predictive cost, but the result demonstrates that a single-variable causal structure can account for more than 60% of the variance in national life satisfaction.

4.4.2.4 SHAP Analysis

SHAP values were computed for XGBoost, the best-performing model on this dataset.

The dominant predictor is *Log_GDP_per_capita*, with a mean absolute SHAP value of approximately 0.39. This is consistent with the BN's identification of GDP as the sole direct parent of *Life_Ladder* and provides convergent evidence from two independent analytical methods that GDP is the primary driver of cross-national variation in life satisfaction. All directional effects are consistent with the World Happiness Report theoretical framework: higher GDP, social support, life expectancy, freedom, and lower perceived corruption all push predictions toward higher life satisfaction. The effects are monotone and theoretically expected without exception, indicating that the XGBoost model has learned a coherent, interpretable representation of the macro-level wellbeing literature.

Social_support is the second most important SHAP predictor, followed by *Healthy_life_expectancy_at_birth* and *Freedom_to_make_life_choices*. Negative affect is a notable contributor in the negative direction, as expected. Generosity is among the weakest predictors by SHAP importance despite its appearance as an upstream node in the BN graph, consistent with the empirical literature suggesting that generosity's relationship to national life satisfaction is indirect and mediated by social capital effects.

4.4.2.5 Sensitivity Analysis

To assess whether positive and negative affect function as proxies for the target - effectively measuring similar evaluative states as life satisfaction itself - the full ML pipeline was re-run after removing both affect variables.

Table 4.11. WHR Gallup - Sensitivity analysis (without affect variables).

TODO table

Model	CV R^2 (full)	CV R^2 (no affect)	ΔR^2
Linear Regression	0.763	0.748	−0.015
XGBoost	0.850	0.832	−0.018

Removing positive and negative affect reduces CV R^2 by only 0.018 for XGBoost and 0.015 for linear regression. This is the smallest sensitivity-induced reduction in this study and indicates that the affect variables are not functioning as dominant proxies for the Cantril scale in the way that *Mental_health_state* and *Gen_health_state* do in the Canadian Survey. Their removal does not materially alter the SHAP importance ranking of the remaining variables: *Log_GDP_per_capita* remains dominant, and the ordering of social support, life expectancy, and freedom is preserved. This stability suggests that the WHR Gallup model's predictive structure is robust and not dependent on the inclusion of evaluatively similar affect measures.

References

[Breiman2001] Breiman, L.: Statistical modeling: The two cultures, Statistical Science, 16(3) (2001), 199–231.

[Lundberg2017] Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions, Advances in Neural Information Processing Systems (NIPS 2017), 30 (2017), 4765–4774.

[Molnar2019] Molnar, C.: Interpretable Machine Learning: A Guide for Making Black Box Models Explainable, Leanpub, 2019.

[OECD2020] Organisation for Economic Co-operation and Development: Measuring well-being and progress, OECD Publishing, Paris, 2020, https://www.oecd.org/en/topics/measuring-well-being-and-progress.html

[Pavot1993] Pavot, W., Diener, E.: Review of the Satisfaction With Life Scale, Psychological Assessment, 5(2) (1993), 164–172.

[Pearl2009] Pearl, J.: Causality: Models, Reasoning, and Inference, 2nd ed., Cambridge University Press, Cambridge, 2009.

[Scutari2014] Scutari, M., Denis, J.-B.: Bayesian Networks: With Examples in R, Chapman & Hall/CRC, Boca Raton, 2014.

[Shmueli2010] Shmueli, G.: To explain or to predict?, Statistical Science, 25(3) (2010), 289–310.

[Yarkoni2017] Yarkoni, T., Westfall, J.: Choosing prediction over explanation in psychology: Lessons from machine learning, Perspectives on Psychological Science, 12(6) (2017), 1100–1122.

[Hastie2009] Hastie, T., Tibshirani, R., Friedman, J.: The Elements of Statistical Learning, 2nd ed., Springer, 2009.

[Kuhn2013] Kuhn, M., Johnson, K.: Applied Predictive Modeling, Springer, 2013.

[Helliwell2023] Helliwell, J.F., Layard, R., Sachs, J.D., De Neve, J.-E.: World Happiness Report 2023, Sustainable Development Solutions Network.

[Diener1984] Diener, E.: Subjective well-being, Psychological Bulletin, 95(3) (1984), 542–575.

[Friedman2001] Friedman, J.H.: Greedy function approximation: A gradient boosting machine, Annals of Statistics, 29(5) (2001), 1189–1232.

[Goldstein2015] Goldstein, A., Kapelner, A., Bleich, J., Pitkin, E.: Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation, Journal of Computational and Graphical Statistics, 2015.

[Morgan2015] Morgan, S.L., Winship, C.: Counterfactuals and Causal Inference, 2nd ed., Cambridge University Press, 2015.